Employee Payroll Program

Aim/Objective:

To develop a Java program to demonstrate the use of an interface and runtime polymorphism for calculating the salary of Regular and Contract employees.

System Requirements:

• Operating System: Windows or any operating system supporting Java.

• Programming Language: Java.

• Java Development Kit (JDK).

• Text Editor or IDE: VS Code / Eclipse / IntelliJ IDEA.

• Command Prompt or Terminal.

Algorithm/Logic Description:

1. Start.
2. Create an interface named Employee with the method calculateSalary().
3. Create the RegularEmployee class and implement the Employee interface.
4. Assign values for Basic Pay, HRA, and T.A for a regular employee.
5. Calculate the total salary by adding Basic Pay, HRA, and T.A.
6. Create the ContractEmployee class and implement the Employee interface.
7. Assign values for Basic Pay and T.A for a contract employee.
8. Set HRA as 0 for the contract employee.
9. Calculate the total salary by adding Basic Pay and T.A.
10. Read the employee type from the user.
11. If the type is Regular, create a RegularEmployee object.
12. If the type is Contract, create a ContractEmployee object.
13. Call calculateSalary() using the Employee reference.
14. If the employee type is invalid, display an error message.
15. Close the Scanner.
16. Stop.

UML Class Diagram:

```
       +----------------------+
       | <<interface>>        |
       |      Employee        |
       +----------------------+
       | calculateSalary():void|
       +----------------------+
              ^          ^
              |          |
implements    |          |    implements
              |          |
+--------------------+  +--------------------+
| RegularEmployee    |  | ContractEmployee   |
+--------------------+  +--------------------+
| basicPay : int     |  | basicPay : int     |
| hra : int          |  | ta : int           |
| ta : int           |  +--------------------+
+--------------------+  | calculateSalary()  |
| calculateSalary()  |  +--------------------+
+--------------------+

       +-------------------------+
       |    EmployeePayroll      |
```

```
                    +------------------------+
                    | main(String[] args)    |
                    +------------------------+
```

Source Code:

```
import java.util.Scanner;

interface Employee {
    void calculateSalary();
}

class RegularEmployee implements Employee {
    int basicPay = 25000;
    int hra = 15000;
    int ta = 5000;

    public void calculateSalary() {
        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

class ContractEmployee implements Employee {
    int basicPay = 12000;
    int ta = 3000;

    public void calculateSalary() {
        int total = basicPay + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: 0");
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

public class EmployeePayroll {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee Type (Regular/Contract): ");
        String type = sc.nextLine();

        Employee emp;

        if (type.equalsIgnoreCase("Regular")) {
            emp = new RegularEmployee();
            emp.calculateSalary();
        }
        else if (type.equalsIgnoreCase("Contract")) {
            emp = new ContractEmployee();
            emp.calculateSalary();
        }
        else {
            System.out.println("Invalid Employee Type");
        }

        sc.close();
    }
}
```

Compilation and Execution Commands:

```
Save the file as: EmployeePayroll.java

Compile:
javac EmployeePayroll.java

Run:
java EmployeePayroll
```

Output Screenshot:

```
javac EmployeePayroll.java
java EmployeePayroll
Enter Employee Type (Regular/Contract): Regular
Salary Details:
Basic Pay: 25000
HRA: 15000
T.A: 5000
Total Amount: 45000
```

Result:

Thus, the Java program was successfully executed and the salary of Regular and Contract employees was calculated using an interface and runtime polymorphism.